MATH 152 **QUIZ 9** NAME _____

Mrs. Bonny Tighe 25 points

11.9 TA_____Wed Wed 11/14/2018

1. Find the power series representation for the following functions.

a) $f(x) = \frac{x}{5 - x^3}$ Find the interval of convergence for this series only .

b) $f(x) = \ln(3x + 1)$

c) $f(x) = \dfrac{x}{(1+x^2)^2}$ Place your final answer here $\sum_{n=0}^{\infty}$ _____

2. Approximate the integral using power series and 4 non-zero terms.

$$\int_0^{0.2} \frac{4x}{1-2x^2}\,dx$$

3. Find $f'(x)$ for $f(x)=\sum_{k=0}^{\infty}\frac{(-1)^k 3^k \quad x^{k+2}}{(1\cdot 5\cdot 9\cdots(4k-3))}$. Then find the radius and interval of convergence.

4. Express the function as the sum of power series by first using partial fractions. Find the interval and radius of convergence. $f(x)=\frac{2+x}{1-4x+3x^2}$

MATH 152 **QUIZ 9A** NAME _____

Mrs. Bonny Tighe 25 points

11.9 TA_____Wed Wed 11/14/2018

1. Find the power series representation for the following functions.

a) $f(x) = \dfrac{9x^2}{1+3x^3}$ Find the interval of convergence for this series only .

b) $f(x) = \ln(x^3 + 4)$

c) $f(x) = \dfrac{x}{(9 - x^2)^2}$ Place your final answer here $\sum_{n=0}^{\infty}$ _____

2. Approximate the integral using power series and 4 non-zero terms.

$$\int_0^{0.2} \frac{4}{1 - 2x^2}\, dx$$

3. Find $f'(x)$ for $f(x)=\sum_{k=0}^{\infty}\frac{(-1)^k 4^k \quad x^{2k}}{(1\cdot 5\cdot 9\cdots(4k-3))}$. Then find the radius and interval of convergence.

4. Express the function as the sum of power series by first using partial fractions. Find the interval and radius of convergence. $f(x)=\frac{1-5x}{1+2x-3x^2}$